

Oct 1 warmup

$$\lim_{n \rightarrow \infty} \frac{5n^5}{e^n} \rightarrow \frac{25n^4}{e^n} \rightarrow \frac{100n^3}{e^n} \rightarrow \frac{300n^2}{e^n} \rightarrow \frac{600n}{e^n} \rightarrow \frac{600}{e^n}$$

$$\lim_{n \rightarrow \infty} \frac{5n^5}{e^n} = \frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{600}{e^n} = \frac{600}{\text{really big}} = 0$$

$$\lim_{n \rightarrow \infty} e^n \approx \infty$$

$$\lim_{n \rightarrow \infty} \frac{f'}{g'}$$

L'Hopital's rule

you can keep reapplying it